Going Agentic: The Right Scaffold at the Right Time

Recap of the Findings ($N = 192$)

Diana Kozachek & Andreas Göldi

University of St. Gallen, Department of Information Systems

diana.kozachek@unisg.ch

Abstract

We created and evaluated an agentic tutoring system that supports concept mapping using four scaffolding mechanisms while adapting intensity and order to pretested knowledge. In a controlled evaluation ($N = 192$), early metacognitive scaffolding produced significantly higher learning gains than a reversed order or no-scaffolding, with the largest gains for learners with lower prior knowledge. We contribute (1) an artifact for mechanism-aware tutoring, (2) empirical evidence that structured scaffolding improves learning only when metacognitive support comes first, and (3) a demonstration that order of delivery is a consequential design variable for intelligent tutoring systems; specifically that scaffolding effects depend on it.

1 Research Questions and Hypotheses

Most Intelligent Tutoring Systems (ITS) vary only help intensity or use a single scaffolding mechanism, leaving open whether mechanism choice and order matter. We focus on four mechanisms, conceptual, procedural, strategic, and metacognitive, and hold intensity adaptive to prior knowledge. We compare two fixed orders of delivery, metacognitive-first and -last against a neutral, as well as a no-scaffolding control condition. Two design uncertainties motivate our research questions. First, neutral, generic agents can provide competent help yet promote cognitive offloading, so it is unclear whether structured scaffolding reliably outperforms them in short sessions [11, 21]. Second, metacognitive prompts can prime planning and monitoring that make later supports more effective, suggesting that position may determine efficacy [10]. These research gaps yield two questions, leading to two hypotheses:

RQ1: *How does structured scaffolding improve learning relative to a neutral agent without scaffolding?*
RQ2: *How does a metacognitive-first or -last order of delivery impact learning gains?*

H1: *There will be significant differences in learning gains when scaffolding is applied vs. regular conversational agent interactions.*

H2: *The metacognitive-first order of delivery condition will produce significantly higher learning gains compared to interactions with a metacognitive-last agent.*

We employed a between-subjects experimental design with three conditions to test the hypothesis that metacognitive-first scaffolding enhances learning performance compared to reversed order or neutral interactions. Participants were randomly assigned via deterministic hash-based randomization: the hash of each participant's unique session identifier was converted to a uniform number and partitioned into conditions, which yielded approximately equal group sizes, fixed each participant's condition across all rounds, and prevented self-selection [2].

2 Procedure

First, we collected learner profiles through self-assessment and a six-item pre-knowledge assessment. Second, participants completed five concept mapping rounds. They began with a baseline round without agentic assistance and followed by four rounds with agentic interaction, in which the map could be expanded. The experimental design required participants to create concept maps. Third, participants completed a six-item post-knowledge assessment and a sixteen-item cognitive load questionnaire developed by Krieglstein et al. [12] for the measurement of three cognitive load types during learning activities. Fourth, we exported session logs for analysis. Fifth, we prepared data for inferential tests.

3 Scaffolding Agents

Metacognitive Scaffolding Agent emphasizes self-regulation and learning process awareness through phase-sensitive adaptation [4]. It supports goal setting and encourages the learner to review their process [1]. In monitoring phases, it facilitates progress assessment and understanding verification. During evaluation phases, it guides learning outcome reflection and strategy effectiveness assessment. The agent maintains categorized reflection prompts [6], explicitly teaches metacognitive strategies, and helps learners recognize their own thinking patterns, as well as mistakes, through structured self-assessment frameworks [14].

Strategic Scaffolding Agent focuses on problem-solving approaches and organizational strategies for concept map development [5]. This agent encourages the learner to set up their own strategic planning protocols and systematic expansion methodologies [15]. If the map is not yet well developed, the agent offers general organizational guidance and next-step recommendations based on current map state. The agent adapts the strategy recommendations based on map complexity and learners' potential to evolve their results, while encouraging their personal strategy deployment [3].

Procedural Scaffolding Agent provides resource- and tool-specific guidance and systematic process support [7]. It addresses common procedural challenges including concept creation, relationship establishment, map organization, and interface navigation [19]. The agent automatically detects procedural issues through map analysis, provides context-sensitive guidance adapted to current round requirements, and implements progressive support reduction as learners gain proficiency with the concept mapping interface [8].

Conceptual Scaffolding Agent targets content accuracy and domain knowledge integration [9]. Operating at three intensity levels, it provides specific concept gap identification and theoretical framework integration at high intensity, general relationship questioning and organizational guidance at medium intensity, and open-ended reflection prompts at low intensity [3]. The agent maintains domain expertise through expert concept map comparison, detects conceptual misconceptions, and provides progressive disclosure of domain insights aligned with learner readiness levels [16].

None of the scaffolding agents gave away suggestions for concrete concepts to add, but encouraged the learner to come up with their own ideas. The distinct behavior of the scaffolding agents are implemented through (1) system messages, priming the connected models to maintain specific roles, including the restrictions. The (2) prompt templates, were co-developed with three educators, who used their domain knowledge to linguistically adapt the tone of the scaffolds [17, 20]. These templates were a crucial component, as they pass on mechanism-specific reply-patterns [13]. And finally, the (3) user input in form of chat messages, which are dynamically assessed in each round and interaction, ensuring context-aware scaffolds.

4 Results

The analysis dataset contained equal group sizes for no scaffolding, metacognition-last support, and metacognition-first support ($n = 64$ each; i.e., $N = 192$). Knowledge gain on the objective test described above was the primary outcome. Metacognition-first support showed the largest mean gain ($M = 0.61$, $SD = 1.48$), as shown in Figure 2. By design, the only manipulated factor was the assigned order-of-delivery; scaffold intensity was adaptive in the same way across scaffolding conditions.

Table 1: Tukey HSD Post-Hoc Test for Knowledge Gains

Group 1	Group 2	Meandiff	p-adj	Lower	Upper	Significant
Neutral	Metacog-first	0.5781	0.0309	0.0425	1.1137	True
Neutral	Metacog-last	0.0312	0.9896	-0.5044	0.5669	False
Metacog-first	Metacog-last	-0.5469	0.0442	-1.0825	-0.0113	True

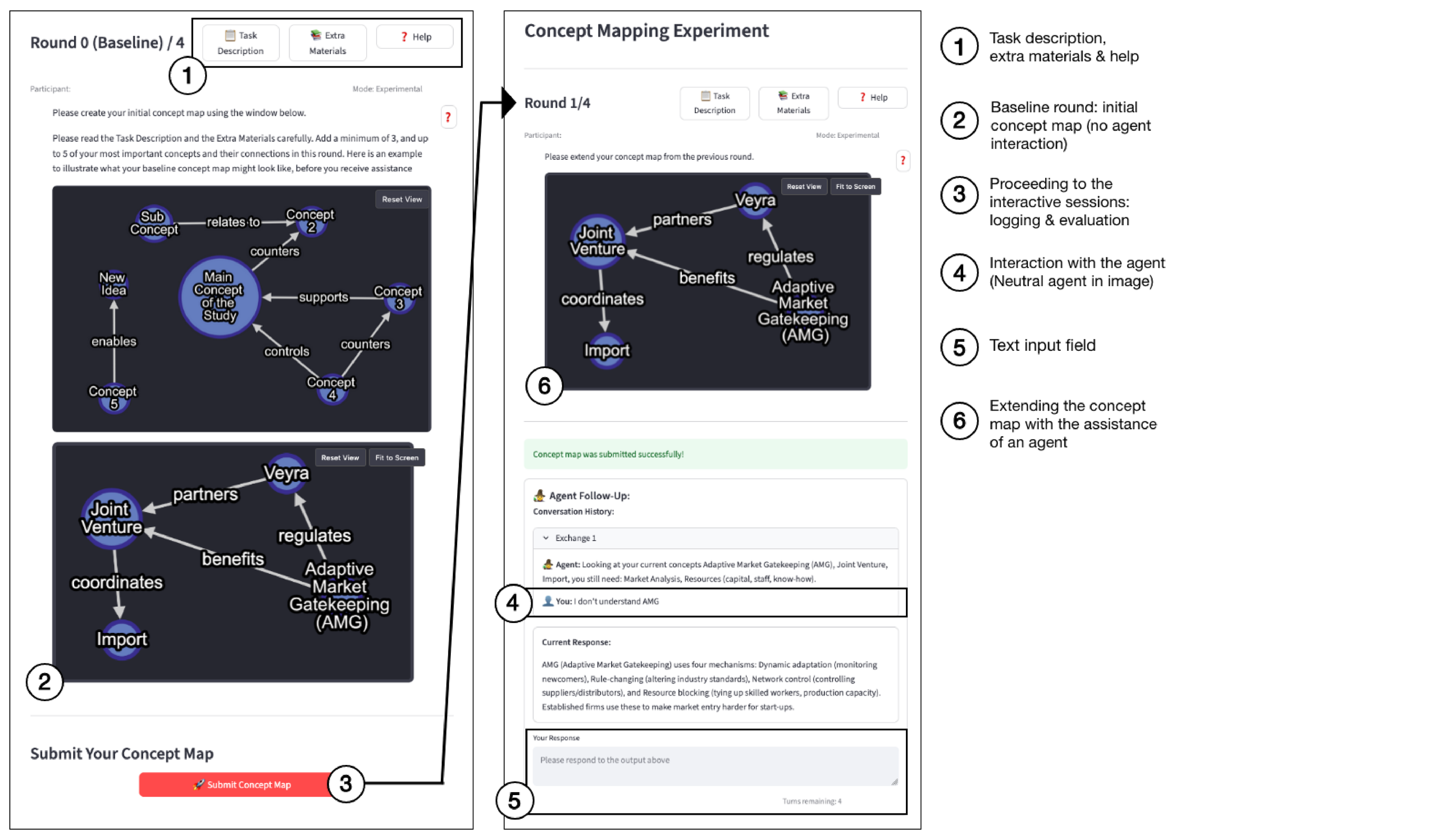

Figure 1: Experimental frontend with neutral chatbot interaction.

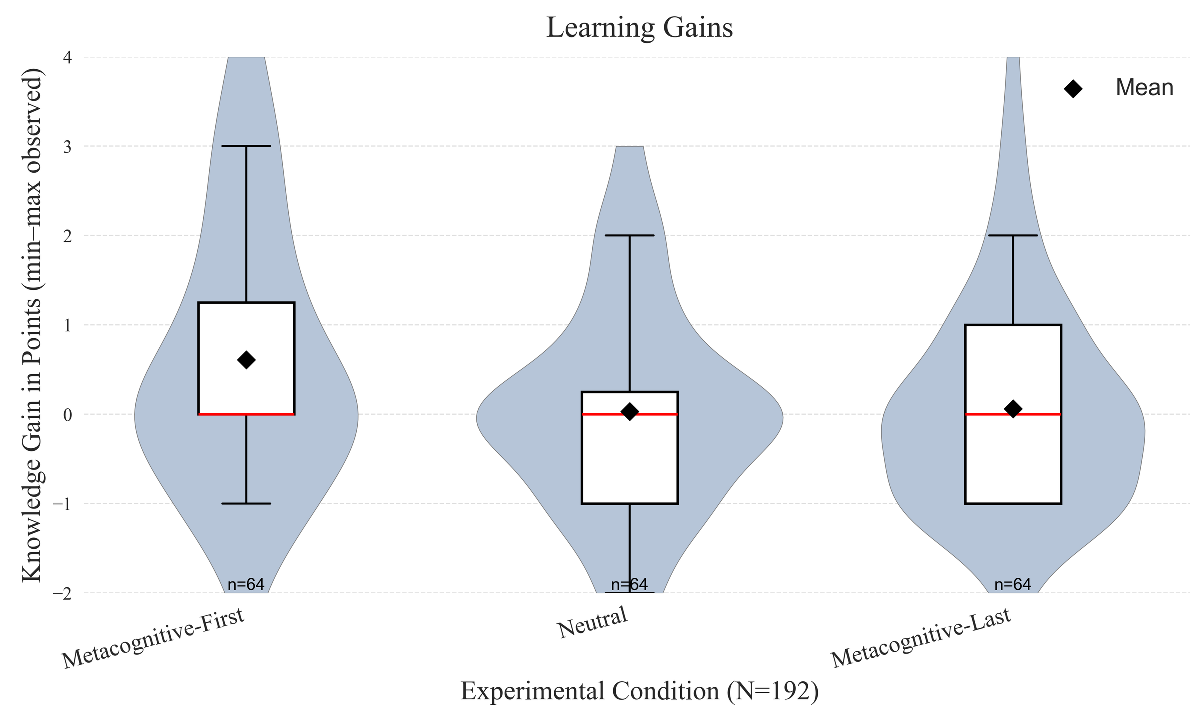

Figure 2: Knowledge gains across experimental conditions. The Y-axis is scaled to the minimum (–0.75) and maximum (3.18) observed knowledge gain in the sample, ensuring that group differences are visible without truncating the data range. Mean $\pm 95\%$ CI.

Conclusion

The evidence maintains **H1** (scaffolding alters learning gains relative to a neutral agent) in a restricted form and maintains **H2** (a metacognitive-first order yields higher gains than metacognitive-last). Specifically, structured scaffolding outperforms a neutral agent only when the *order of delivery* places metacognitive support first, as shown by the between-group contrasts and the within-group improvement for the metacognitive-first condition. Mechanism differentiation without this ordering did not exceed the neutral baseline, which rejects the broader reading that any structured help suffices, consistent with theory that early metacognitive prompts prime effective strategy use [18, 10].

Mechanism-aware agentic ITS is practical and testable and operationalizes conceptual, procedural, strategic, and metacognitive scaffolds with adaptive intensity tied to concept-mapping diagnostics. This design turns educational constructs into executable policies that a system can apply consistently during interaction.

Providing metacognitive scaffolding first improves learning only when metacognitive support comes first. This contribution specifies the condition under which mechanism-aware scaffolding yields benefits, aligning with accounts that planning and monitoring enable productive use of later supports.

Order of delivery is a primary design parameter is a primary design parameter, independent of help intensity. This contribution shifts attention from how much help to when and which help, a distinction emphasized in instructional theory.

Limitations and Next Steps

Deploy Longitudal Study: single-session design cannot address whether metacognitive-first advantages persist across extended learning periods or fade as learners develop expertise. We need to assess learning over the course of multiple sessions.

Testing True Adaptability: by coupling learning diagnostics with policy selection that updates order in real time, for example gating strategic or conceptual supports on metacognitive readiness, or multiple scaffold mechanisms combined.

Assessing Replicability: due to the system logs, concept-map trajectories and conversations, we can re-run evaluations on identical inputs.

Improving Learning Diagnostics: real-time data pipelines shall stabilize responses and agent coordination by adding runtime guardrails, validation checks on agent outputs, and more expressive diagnostics that trigger dynamic scaffold adjustments, all without altering the scaffold definitions.

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